

Report on AARC Student Attendance and Academic Performance

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Academic Assistance and Resource Center (AARC) Analysis

Prepared from the student attendance analysis output

Executive Summary

This report summarizes an analysis of student attendance records from the Academic Assistance and Resource Center (AARC) at Stephen F. Austin State University. The purpose of the analysis was to examine how students use AARC services, identify attendance patterns across centers and courses, and explore whether AARC attendance is associated with student academic performance.

The dataset contained 91,900 visit records, 8,891 unique students, 7 academic terms, 1,652 courses, and 5 AARC centers. The analysis suggests that AARC engagement is positively associated with student academic performance, especially when comparing average attendance across grade levels. However, the correlations are weak to moderate, so the findings should be interpreted as evidence of association rather than proof of causation.

The most important practical finding is that students with higher grades tended to have more AARC visits and more total time in AARC services. Students earning A grades averaged about 16 visits, while students in the F/W category averaged about 3 visits. This suggests that consistent use of academic support is linked with stronger academic outcomes.

1. Introduction

The Academic Assistance and Resource Center (AARC) provides academic support services to students through tutoring, supplemental instruction, writing support, workshops, and other learning assistance programs. This report analyzes student attendance in AARC services and connects attendance patterns with final course grades.

The central question guiding the analysis is: How is AARC attendance related to student academic performance? The report also identifies which centers and courses had the highest demand, how visit duration varied, and what the observed correlations mean in practical terms.

2. Data Description and Preparation

The analysis used AARC attendance data containing **student identifiers, term, course, visit duration, check-in date and time, center attended, reason for visit, and final course grade.**

Data Component	Value
Visit records	91,900
Unique students	8,891
Academic terms	7
Courses	1,652
AARC centers	5

Final course grades were converted into numeric values so they could be used in correlation analysis and grouped comparisons. A was coded as 4, B as 3, C as 2, D as 1, and F/W as 0.

Grade	Numeric Value
A	4
B	3
C	2
D	1
F / W	0

Grade Distribution

Grade	Count
A	28,096
B	26,914
C	21,749
D	6,102
F	4,858
W	4,181

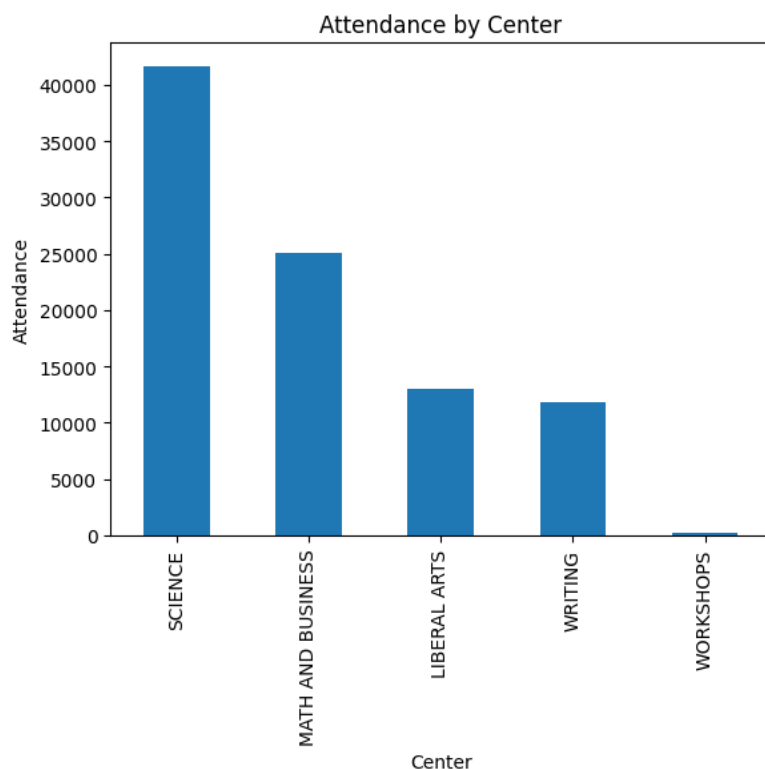
Most AARC attendance records were associated with students who eventually earned A, B, or C grades, while fewer records were associated with D, F, or withdrawal outcomes. This indicates that AARC is used by students across performance levels, not only by students who are failing.

3. Attendance Patterns by AARC Center

Center	Total Visits
Science	41,661
Math and Business	25,143
Liberal Arts	13,034
Writing	11,863
Workshops	199

The Science center received the largest share of AARC traffic, followed by Math and Business. This suggests strong demand for support in STEM and quantitative courses. These areas may include courses with higher difficulty, large enrollments, or strong supplemental instruction participation.

Workshops had very low attendance compared with the other centers. This does not necessarily mean workshops were ineffective. It may mean they were offered less often, served smaller groups, or functioned differently from regular tutoring and SI sessions.

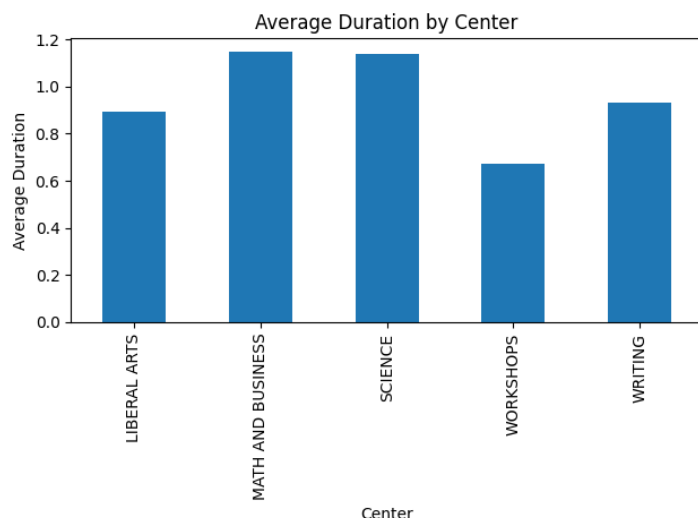


4. Average Visit Duration by Center

Center	Average Duration
Math and Business	1.146
Science	1.138
Writing	0.932
Liberal Arts	0.895
Workshops	0.670

Math and Business and Science had the longest average visit durations. Practically, this may indicate that students in these areas require longer sessions because the material often involves calculations, problem-solving, laboratory concepts, or step-by-step explanation.

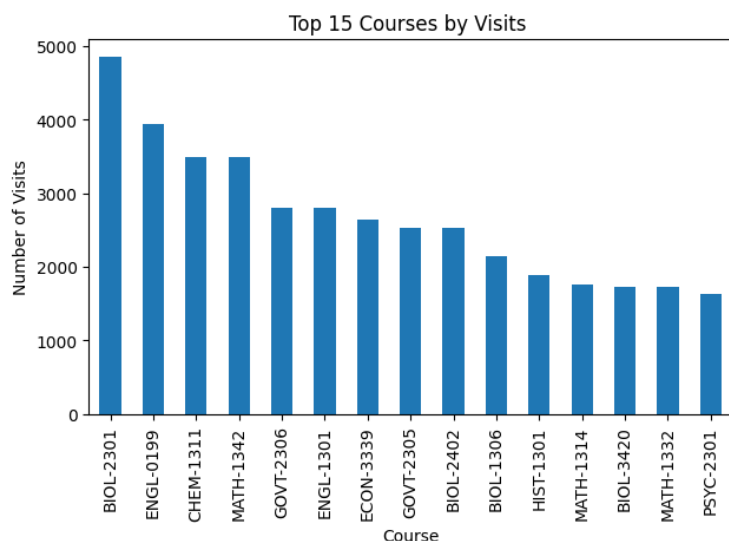
Writing and Liberal Arts had shorter average durations, which may reflect the type of assistance provided, such as reviewing drafts, discussing assignments, or clarifying concepts. Workshop recorded the least average duration, which may be due the fact that they were offered less often, or functioned differently from regular tutoring sessions.



5. Most Frequently visited Courses

Course	Visits
BIOL-2301	4,846
ENG-0199	3,939
CHEM-1311	3,497
MATH-1342	3,487
GOVT-2306	2,803

Biology courses appeared frequently among the most visited courses. This suggests that Biology is one of the major areas where students seek academic support through the AARC. High numbers of visits may reflect course difficulty, large enrollment, or the availability and popularity of support for those courses.



6. Student-Level Attendance and Grade Relationship

The student-level analysis grouped records by student ID and calculated each student's total number of AARC visits, total duration of AARC visits, and highest recorded numeric grade.

Variables Compared	Correlation
Total visits and total duration	0.9467
Total visits and grade	0.2945
Total duration and grade	0.2608

6.1 Total Visits and Total Duration: $r = 0.9467$

This is a very strong positive correlation. Students who visited the AARC more often also spent more total time in AARC services. This is expected because more visits naturally increase total duration.

Practical meaning: visit count and total duration are measuring very similar aspects of student engagement. A student with many visits is usually also a student with many total hours of support. This correlation does not show that more time causes better grades; it mainly confirms that visit frequency and total time are strongly connected.

6.2 Total Visits and Grade: $r = 0.2945$

This is a positive but weak-to-moderate correlation. Students with more AARC visits tended to have higher grades, but the relationship is not strong enough to say that attendance alone determines academic success.

Practical meaning: AARC attendance may be associated with better academic outcomes. Students who repeatedly use AARC services may benefit from continued academic support, clarification of course material, and structured study assistance. However, other factors also play important roles, including motivation, prior preparation, course difficulty, instructor differences, study habits, major, workload, and whether students sought help early or only after struggling.

6.3 Total Duration and Grade: $r = 0.2608$

This is also a positive but weak-to-moderate correlation. Students who spent more total time in AARC services tended to earn slightly higher grades.

Practical meaning: greater time investment in academic support is associated with better performance, but the relationship is not strong enough to guarantee that long AARC attendance leads to high grades. Some students may spend many hours in AARC because they are already struggling, while others may perform well with fewer visits because they already have strong preparation.

7. Average Attendance by Grade Level

Numeric Grade	Letter Interpretation	Average Visits	Average Total Duration
0	F/W	3.27	3.55
1	D	4.30	4.72
2	C	6.50	6.84
3	B	9.68	10.70
4	A	16.10	17.21

This is one of the clearest findings in the analysis. Students with higher grades had higher average AARC attendance and higher average total duration.

Students who earned A grades averaged about 16 visits, while students in the F/W category averaged only about 3 visits. Practically, this suggests that students who engage more consistently with AARC services are more likely to be associated with stronger academic outcomes. However, this should not be interpreted as proof that AARC attendance alone caused the higher grades. High-performing students may also be more proactive and may use AARC early and consistently.

8. Center-Level Attendance and Average Grade

Center	Total Visits	Average Grade
Liberal Arts	13,034	2.849
Math and Business	25,143	2.349
Science	41,661	2.825
Workshops	199	3.302
Writing	11,863	2.377

The correlation between total center visits and average grade was -0.3676. This is a moderate negative correlation.

Practical meaning: centers with more visits were not necessarily the centers with higher average grades. This does not mean that high-attendance centers are ineffective. A more reasonable interpretation is that heavily

used centers may serve more difficult courses or students who are already at greater academic risk. For example, Science and Math and Business may attract many students from challenging gateway courses.

9. Biology Course Analysis

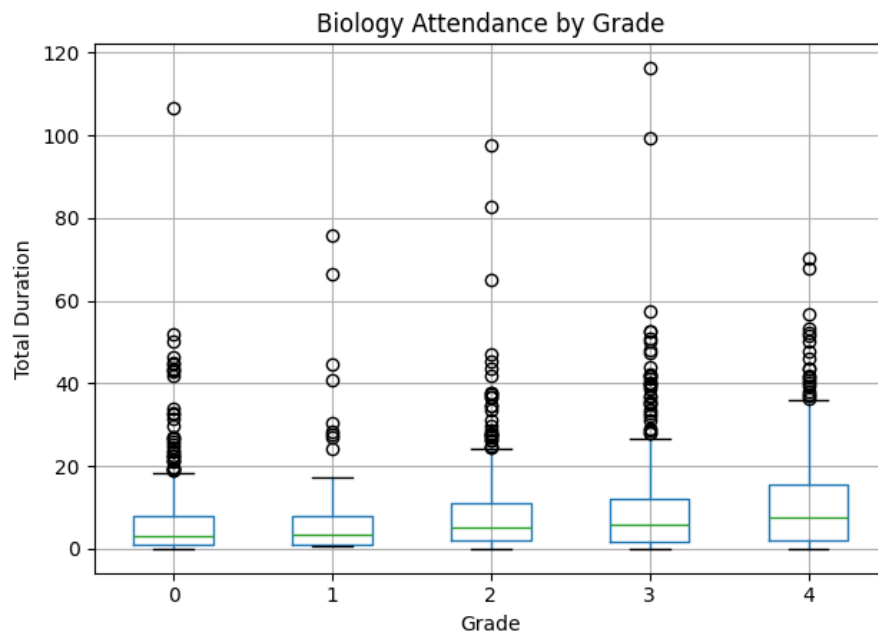
Because Biology courses were among the most frequently visited courses, a separate analysis was performed for Biology-related attendance.

Numeric Grade	Average Visits	Average Total Duration
0	5.77	6.81
1	7.09	7.88
2	7.75	8.78
3	8.77	9.20
4	10.81	10.86

Variables Compared	Correlation
Total Biology visits and grade	0.1789
Total Biology duration and grade	0.1208

For Biology courses, students with higher attendance tended to have slightly higher grades, but the relationship was weak. The positive correlation of 0.1789 between Biology visits and grade means that Biology students who attended more AARC sessions generally performed a little better, but the relationship was not strong.

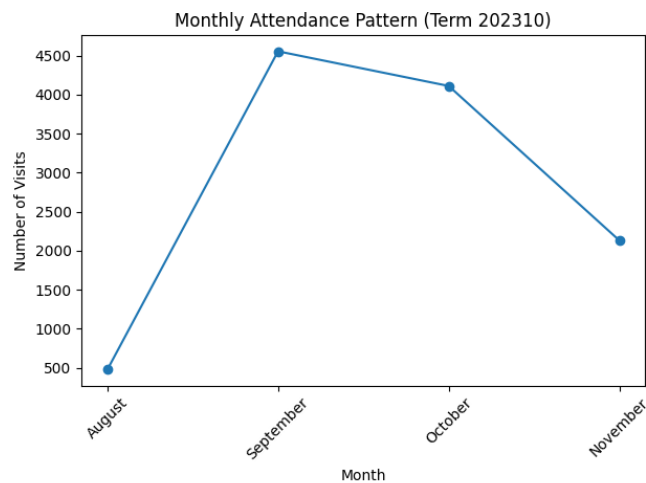
The positive correlation of 0.1208 between Biology duration and grade means that spending more total time in Biology-related AARC support was also slightly associated with higher grades. Practically, this means AARC attendance may help Biology students, but attendance alone does not explain Biology performance. Other factors such as prior science background, exam preparation, lab performance, course section, instructor expectations, and timing of the first visit likely matter. Also, some high-performing students don't need many visits.

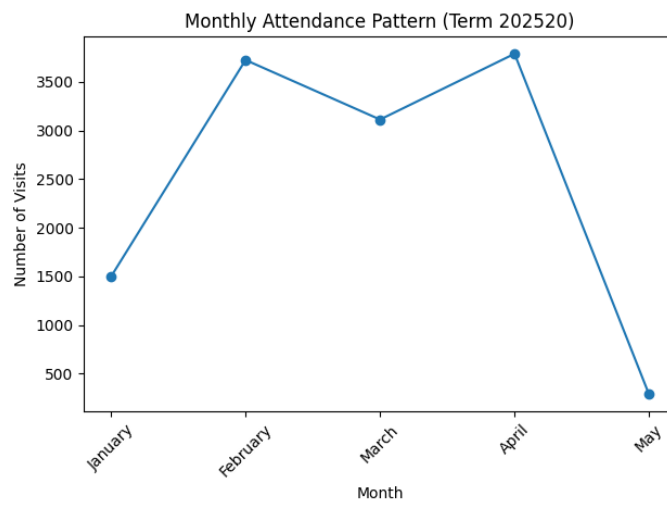


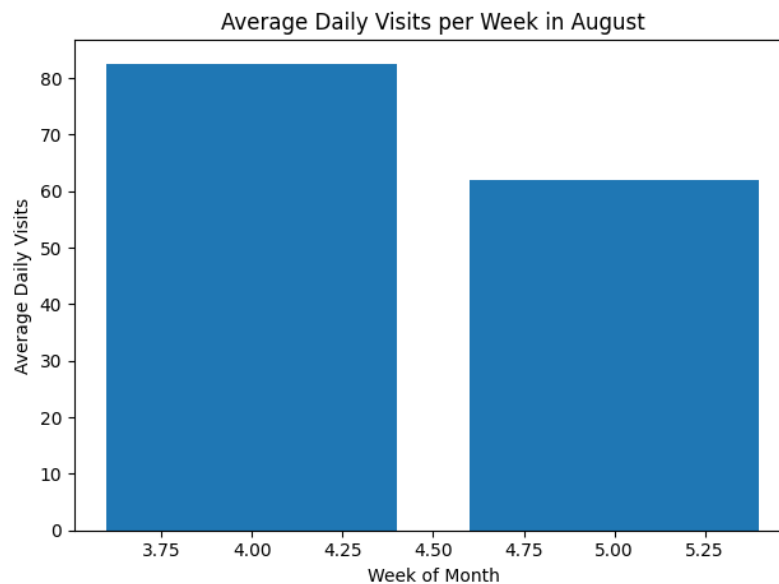
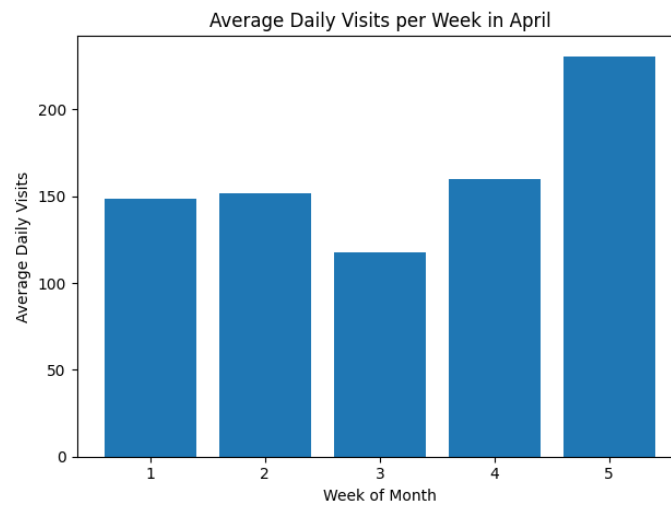
10. Peak Attendance Patterns

The analysis also examined attendance over time by month and week of the month. Attendance varied across academic terms and appeared to rise during academically intense periods, such as after the semester begins and around exam-heavy months.

Practical meaning: AARC demand is not constant throughout the semester. Attendance tends to increase during periods when assignments, tests, and course pressure increase.







11. Main Findings

- AARC usage was highest in the Science center, followed by Math and Business, showing strong demand for STEM and quantitative course support.
- Students with higher grades tended to have higher average AARC attendance. A-grade students averaged about 16 visits, while F/W students averaged about 3 visits.
- Student-level correlations were positive. Total visits and grades had a correlation of 0.2945, while total duration and grades had a correlation of 0.2608. This means AARC engagement is associated with better performance, although the relationship is not strong enough to claim that attendance alone determines grades.

- Visit count and total duration had a very strong correlation of 0.9467, meaning that students who visit more often also spend more total time in AARC support.
- Biology-specific correlations were positive but weak. Biology visits and grades had a correlation of 0.1789, while Biology duration and grades had a correlation of 0.1208.
- Center-level analysis showed a negative correlation between total center visits and average center grade. This likely reflects the fact that high-attendance centers may serve more difficult courses or students who are already struggling.

12. Recommendations

Encourage earlier AARC attendance. Students may benefit more if they begin attending AARC services early in the semester rather than waiting until they are struggling.

Strengthen support for high-demand centers. Science and Math and Business had the highest attendance, so these areas may need strong staffing, more tutors, and extended support during peak periods.

Monitor high-risk courses. Biology courses appeared frequently among the most visited courses. The AARC may consider targeted support for high-demand Biology courses such as BIOL-2301, BIOL-3420, and BIOL-2402.

Use attendance data with caution. Attendance is positively related to grades, but it is not the only factor that explains academic success. Future analyses should include prior GPA, classification, major, course difficulty, and timing of first AARC visit.

Focus on timing and consistency. Future analysis should examine whether students who attend early and consistently perform better than students who attend only near exams or after falling behind.

13. Limitations

This analysis shows associations, not causation. A positive correlation between AARC attendance and grades does not prove that AARC attendance caused higher grades.

Some students may attend AARC because they are already struggling, which can weaken the observed relationship between attendance and grades. Also, high-performing students may attend AARC proactively, while others may attend only after academic difficulty has already developed.

14. Conclusion

Overall, the analysis suggests that AARC attendance is positively associated with student academic performance. Students who visited more often and spent more time in AARC services tended to earn higher grades. This pattern was especially clear when comparing average attendance across grade levels.

However, the correlations were generally weak to moderate, meaning that AARC attendance is only one factor related to academic success. The findings support the value of AARC services, but they also show that future analysis should consider additional student and course-level factors.

The practical message is that AARC engagement appears beneficial, especially when students use it consistently, but attendance data should be interpreted as part of a broader academic support system rather than as the sole explanation for student grades.